

Scenario 1 · Medical triage

A neural network turns vital signs into an urgency class

Facts on the table. A hospital trust wants to replace its rule-based early-warning score with a learned model. In this afternoon’s exercise the 96 %-accurate model under-triaged a patient with SpO₂ 86 % (found in 0.5 s); the corrected model still failed at exactly 90 mmHg; the model trained with the properties in the loop proved all five clinical limits but over-triages at the thresholds (92 % agreement with the strict rule). Elsewhere: a certified neural dosing controller for vancomycin was proved never to exceed the safe concentration over an unbounded sequence of doses — under a simplified pharmacokinetic model and synthetic training data (Sirman et al., SAIV 2026). An analysis reported in *Biomedical Safety & Standards* (Feb 2026) attributes nearly half of recent FDA medical-device recalls to AI-enabled devices, mostly for diagnostic or measurement error. Under Article 15 of the EU AI Act (obligations from 2 August 2026) high-risk systems must reach “an appropriate level of accuracy, robustness, and cybersecurity”.

Regulator

ROLE CARD ✂ CUT ALONG THE DASHED LINE

Your goal. Approve only what can be shown safe for every patient the device will see, and be able to explain the approval later.

You must insist on. Written properties with named regions; evidence for *all* inputs in scope, not a test set; a plan for what happens when a counterexample is found after deployment.

The uncomfortable question you ask. *Which clinical rule did you decide not to prove, and why?*

Vendor

ROLE CARD ✂ CUT ALONG THE DASHED LINE

Your goal. Ship a model that is accurate, fast, and defensible; keep the guarantee without losing the accuracy that sold it.

You must insist on. A clear statement of the operating region so that out-of-region inputs are not counted as failures; margins that do not make the product useless.

The uncomfortable question you ask. *Do you want the 96 % model with a hole or the 92 % model with a proof? You cannot have both today.*

Clinician / patient

ROLE CARD ✂ CUT ALONG THE DASHED LINE

Your goal. Never miss the deteriorating patient; not be drowned in false alarms either.

You must insist on. Over-triage has a cost (beds, attention, alarm fatigue) that must be on the sheet; a human can always override; the thresholds are clinical decisions, not engineering ones.

The uncomfortable question you ask. *Who told the model that 92 % saturation is “fine”?*

Verification engineer

ROLE CARD ✂ CUT ALONG THE DASHED LINE

Your goal. Say precisely what has and has not been proved, and refuse to let a proof be quoted beyond its region.

You must insist on. Inputs must be bounded; flags pinned; every threshold a unit away from where the labels change; every counterexample reproduced and root-caused.

The uncomfortable question you ask. *The proof says nothing about a patient whose readings are wrong. What does?*

DECISION SHEET	
The property, in one sentence	
The region (inputs, ranges, units)	
ε / margin and who chose it	
Acceptable failure rate (and for whom)	
Action on a counterexample (in development / in production)	
Who signs, who is accountable when the specification was wrong	
Out of scope for verification and how it is addressed instead	

Scenario 2 · Authentication

Live facial recognition decides who gets stopped

Facts on the table. According to figures reported by PA in 2026, the Metropolitan Police used live facial recognition 231 times in 2025 and scanned about 4.2 million faces; ten people were falsely alerted, eight of them Black; none of the ten was arrested. Shaun Thompson, an anti-knife-crime youth worker, was stopped and detained on a false match in 2024 — his passport and bank cards were “not enough”; on 21 April 2026 the High Court dismissed his and Big Brother Watch’s challenge and ruled the deployment lawful (The Register, 22 April 2026). Industry reports in 2026 describe deepfake camera injection and voice cloning sold as a service against biometric onboarding (vendor sources; treat the figures with care). Robustness of a face-matching network can be stated as a property around known inputs; “never misidentifies a person” cannot.

ROLE CARD

CUT ALONG THE DASHED LINE

Regulator

Your goal. Permit only deployments whose error behaviour is specified, measured and reviewable, with equal protection across groups.

You must insist on. A stated region (lighting, pose, resolution) and ϵ ; error rates *per group*, not overall; what a false alert leads to, in minutes and in law.

The uncomfortable question you ask. *Ten in four million sounds small. Ten of whom?*

ROLE CARD

CUT ALONG THE DASHED LINE

Vendor

Your goal. Deliver a system that catches wanted people and passes an audit.

You must insist on. The operating envelope the numbers were measured in; the threshold that trades false alerts against misses is set by the customer, not the vendor.

The uncomfortable question you ask. *If you demand zero false alerts, you are asking for a system that alerts on nobody. Which do you want?*

ROLE CARD

CUT ALONG THE DASHED LINE

Affected person

Your goal. Not be stopped, filmed and doubted because of a similarity score.

You must insist on. A counterexample here is a person; the “action on a counterexample” must be a five-minute check, never an arrest on the score alone; a way to contest the match.

The uncomfortable question you ask. *What would have convinced the officer that the machine was wrong?*

ROLE CARD

CUT ALONG THE DASHED LINE

Verification engineer

Your goal. State exactly what robustness around a reference image does and does not cover.

You must insist on. An adversary who *chooses* the perturbation (glasses, make-up, injection) is stronger than random noise; the property must say which adversary it assumes.

The uncomfortable question you ask. *Robust to what, chosen by whom?*

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Scenario 3 · Autonomous control

A safety envelope you can prove versus a perception system you cannot

Facts on the table. On 20 April 2026 an empty Waymo robotaxi drove into a flooded road in San Antonio and was swept into a creek; the fleet’s software was written to *slow, but not stop* at water it judged “potentially untraversable”; 3,791 vehicles were recalled and fixed over the air (CNBC, 12 May 2026). In March 2026 the US regulator upgraded its investigation of Tesla’s Full Self-Driving to an engineering analysis over crashes in reduced visibility, covering about 3.2 million vehicles. In the tutorial you saw a neural controller *proved* to keep a car on the road under crosswind and sensor noise (Vehicle + Marabou, with a machine-checked proof of the whole system in Agda), and the ACAS Xu collision-avoidance network *disproved* for Property 3 in under a second.

Regulator

ROLE CARD✂ CUT ALONG THE DASHED LINE

Your goal. Approve a system whose safety case rests on evidence about every situation in the operating domain, and know who to hold responsible.

You must insist on. A written operating domain; proofs for the control layer; an honest account of the perception layer that cannot be proved; incident reporting on time.

The uncomfortable question you ask. *Whose signature is on the sentence “slow, but do not stop”?*

Vendor

ROLE CARD✂ CUT ALONG THE DASHED LINE

Your goal. Deploy widely, fix fast, keep the fleet running.

You must insist on. Verification of the controller is affordable and should be in the pipeline; perception is monitored statistically; over-the-air fixes are part of the safety case, not an admission.

The uncomfortable question you ask. *A recall in ten days is the system working. What would you have preferred?*

Road user

ROLE CARD✂ CUT ALONG THE DASHED LINE

Your goal. Not be the counterexample.

You must insist on. The difference between “the car never leaves the road” (a proof about a controller) and “the car never hits a child” (not a property of pixels) must be explained in plain words before deployment; a human must be able to stop the vehicle.

The uncomfortable question you ask. *Which of the two sentences are you actually promising me?*

Verification engineer

ROLE CARD✂ CUT ALONG THE DASHED LINE

Your goal. Prove what can be proved and label the rest.

You must insist on. Controllers and envelopes: prove them, in the units of the problem. Perception: bound it with robustness properties around known scenes and say so. Specifications are reviewed like code — they were the bug in San Antonio.

The uncomfortable question you ask. *The proof of the controller assumes the sensor is right. Who owns that assumption?*

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